

Deep Learning-Based Synthetic-CT Generation from MRI for Enhanced Precision in MRI-Only Radiotherapy Dose Planning

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Abstract

Introduction: Radiotherapy aims to precisely target tumors while sparing healthy tissue, traditionally relying on CT imaging for accurate dose planning. However, CT has limitations in soft tissue contrast and exposes patients to ionizing radiation. MRI offers superior soft tissue contrast without radiation but lacks electron density information, restricting its use in dose planning. This study addresses this gap by developing deep learning models to generate pseudo-CT images from MRI, enabling MRI-only workflows in radiotherapy.

Methodology: Paired MRI and CT scans from 12 subjects were processed using normalization, alignment, and masking. Four deep learning architectures (U-Net, Pix2Pix, CycleGAN, and conditional GAN (cGAN)) were trained to generate synthetic CT images from MRI data. Model performance was evaluated using metrics including mean absolute error (MAE), mean squared error (MSE), peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and Pearson correlation coefficient (PCC).

Results: Pix2Pix achieved the highest SSIM and PSNR, indicating strong structural preservation and reduced noise. It also had the lowest MAE and MSE, showing high accuracy in synthetic-CT generation. The cGAN model scored highest in PCC, highlighting its effective intensity alignment with real CT data. Statistical tests confirmed Pix2Pix's superior performance, though CycleGAN and cGAN also showed notable results in alignment accuracy.

Conclusion: Deep learning models, particularly Pix2Pix, can generate reliable pseudo-CT images from MRI, supporting MRI-only radiotherapy planning. This approach reduces radiation exposure and may streamline radiotherapy workflows, offering a promising advance for patient-centered cancer care and MRI-only radiotherapy workflows.

Keywords: MRI-only radiotherapy, Deep learning, Synthetic CT images, Cancer treatment planning.

Introduction

Radiotherapy is an essential treatment modality in cancer care, aiming to deliver precise doses of radiation to malignant tissues while sparing surrounding healthy structures. The primary goal of radiotherapy is to achieve maximum tumor control while minimizing damage to adjacent normal tissues. Successful radiotherapy treatment relies heavily on accurate dose planning, typically achieved through precise imaging techniques¹. Computed tomography (CT) imaging has traditionally been the standard in radiotherapy planning due to its capacity to provide accurate electron density information

and detailed tissue contrast, which are essential for calculating dose distributions. However, CT imaging has inherent limitations, including exposure to ionizing radiation and suboptimal soft tissue contrast, which can be particularly challenging when planning treatments near sensitive or complex anatomical structures². In recent years, magnetic resonance imaging (MRI) has emerged as a complementary imaging modality due to its superior soft tissue contrast and non-ionizing nature. Current MRI sequences and protocols can be used to delineate soft tissue structures more clearly than CT makes it valuable in distinguishing tumors from surrounding normal tissues, which is crucial for precise dose targeting. Unlike CT, how-

ever, MRI does not provide electron density information, a key factor in dose calculations. This absence of electron density data has limited MRI's application as a standalone tool in radiotherapy dose planning, necessitating the acquisition of CT images as an additional step to gather this information. This dual imaging approach not only increases patient exposure to radiation but also incurs higher healthcare costs and operational complexity^{2,3}. To overcome these limitations, research has increasingly focused on developing MRI-only workflows for radiotherapy, which eliminate the need for additional CT scans by enabling MRI to serve as the sole imaging modality. However, achieving this goal requires new techniques that can accurately derive electron density information from MRI data. Traditional methods, such as bulk density overrides and voxel-based approaches, have been explored but fall short of providing the precision required for complex dose calculations. This gap has spurred the exploration of advanced techniques, particularly within the field of artificial intelligence (AI), to enhance MRI's utility in radiotherapy⁴. Recent advancements in deep learning have introduced promising solutions to bridge the gap between MRI and CT imaging in radiotherapy. Deep learning models, particularly convolutional neural networks (CNNs) and generative adversarial networks (GANs), have shown remarkable capability in synthesizing pseudo-CT images from MRI data. These pseudo-CT images effectively emulate the electron density information in CT, making it possible to use MRI data alone for dose calculations. By leveraging the strengths of MRI and augmenting them with pseudo-CT synthesis, deep learning offers a pathway to improve the accuracy and efficiency of radiotherapy dose planning. This approach not only reduces radiation exposure but also streamlines the imaging process, potentially lowering healthcare costs and improving patient comfort⁵⁻⁷. The generation of pseudo-CT images from MRI is a complex task that requires high levels of accuracy to ensure safe and effective treatment. Deep learning algorithms have become powerful tools in this area due to their ability to learn complex mappings from large datasets and adapt to variations in anatomical and imaging features. Among the various deep learning models explored, CNNs have been effective for image-to-image translation tasks due to their capacity to capture spatial hierarchies in image data. More recently, GANs and their conditional variants (cGANs) have further enhanced performance in synthetic image generation by introducing a discriminative component that encourages the generation of more realistic images. These deep learning techniques, particularly GANs, can generate high-quality pseudo-CT images that closely resemble actual CT scans in terms of both intensity values and structural integrity, making them highly suitable for radiotherapy applications^{7,8}. This study builds on these advancements by aiming to develop and validate a deep learning model tailored for synthesizing high-fidelity pseudo-CT images from MRI data. The approach involves training models on paired MRI and CT data to learn the complex mappings necessary for accurate electron density estimation. The proposed model leverages a combination of U-Net architecture, conditional GAN (cGAN), Pix2Pix, and

CycleGAN frameworks, each with specific architectural features optimized for generating high-quality synthetic CT images. These models are designed to capture both global and fine-grained anatomical details, which are critical for accurate radiotherapy dose planning. By comparing multiple models, this study seeks to identify the most effective architecture for translating MRI to pseudo-CT images, aiming to achieve a balance between high accuracy and computational efficiency^{9,10}. The implications of successfully integrating pseudo-CT images into radiotherapy workflows are significant. An MRI-only approach would reduce the dependence on CT scans, lowering both patient radiation exposure and healthcare costs. Moreover, MRI-based workflows could streamline clinical protocols, reducing the time required for patient preparation and image acquisition, which would be particularly beneficial in high-demand settings. The potential to generate pseudo-CT images directly from MRI data also aligns with recent trends in precision medicine, where individualized treatment planning based on high-resolution, modality-specific imaging data is increasingly emphasized. In summary, this study's objective is to develop and validate a deep learning algorithm capable of generating high-quality pseudo-CT images from MRI data for radiotherapy dose planning. This innovative approach aims to improve treatment precision by enhancing the accuracy of target delineation and dose distribution, ultimately contributing to safer and more effective cancer treatments. The integration of pseudo-CT images into radiotherapy workflows has the potential to transform clinical practice by reducing the dependency on CT, improving patient outcomes, and lowering healthcare costs, paving the way for a more efficient and patient-centered approach to cancer care^{4,5,11}. This study aimed to develop a deep learning-based method for generating pseudo-CT images from MRI data to enhance radiotherapy dose planning by integrating MRI's superior soft tissue contrast with CT's electron density information.

Methodology

The methodology comprises data acquisition, preprocessing, model architecture, and evaluation.

Materials

This study used a Toshiba Vantage 1.5T MRI machine and a GE Medical System CT scanner to acquire MRI and CT data, respectively. Python was the primary programming language, with supporting libraries like TensorFlow, NumPy, Scikit-learn, and OpenCV for data processing and model implementation. dcm2niix software was used for DICOM-to-NIfTI conversion, and ITK-SNAP provided visualization capabilities.

Data Acquisition

The data for this study were acquired following imaging protocols at Korle Bu Teaching Hospital and other collaborating centers. A total of 12 subjects contributed paired MRI and CT scans focused on the head and neck region, a critical anatomi-

cal area for radiotherapy dose planning. The dataset did not include patients with visible tumors or gross anatomical abnormalities, as the primary objective was to assess baseline model performance. MRI images were acquired using a Toshiba Vantage 1.5T scanner with an in-plane resolution of $1 \times 1 \text{ mm}^2$ and slice thicknesses ranging from 2 mm to 8 mm. The MRI images were acquired using a standard T1-weighted spin echo sequence, selected for its anatomical clarity and widespread clinical availability. No specialized MRI sequences (e.g., Dixon, UTE, or ZTE) were used to ensure clinical relevance and generalizability of the synthetic CT generation process. Corresponding CT scans were obtained using a GE Medical System CT scanner with a resolution of $0.98 \times 0.98 \text{ mm}^2$ and slice thicknesses between 1 mm and 2.5 mm. The dataset was partitioned into 9 cases for training, 1 for validation, and 2 for testing, ensuring diversity in scan parameters and anatomical features across the sets. The MRI and CT scans were acquired 1 to 2 days apart for each subject under standard diagnostic workflows. Although radiotherapy-specific immobilization was not used, both scans were performed in a supine position with neutral head alignment. Minor anatomical differences were noted between modalities, particularly in soft tissues near air cavities. Rigid image registration using mutual information and visual inspection helped minimize these discrepancies, which were typically within 2–3 mm. These small differences may have contributed to minor spatial inconsistencies but did not significantly affect overall model performance.

Preprocessing Pipeline

Before training, the MRI and CT images were processed to standardize and enhance image quality for optimal model performance. First, N4 bias correction was applied to correct for intensity inhomogeneities, a common artifact caused by non-uniform magnetic fields, especially in older MRI scanners¹². This correction minimized low-frequency bias fields that could affect model accuracy. In addition, quality assurance steps were undertaken to minimize MRI geometric distortion. The MRI scanner underwent routine QA using a spatial distortion phantom, ensuring geometric accuracy within $\pm 2 \text{ mm}$. A T1-weighted spin echo sequence, which is less prone to distortion, was used. During preprocessing, rigid registration with CT was also visually inspected to confirm anatomical consistency. No significant non-linear distortions were observed. Next, images were converted to the NIfTI format and anonymized to protect patient confidentiality. To ensure accurate spatial correspondence between modalities, we first resampled the MRI volumes to match the voxel dimensions of the CT scans. This was followed by rigid image registration using SimpleITK, with mutual information as the similarity metric and Euler transformation (rotation + translation) as the transformation model. This step accounted for positional differences due to patient setup variation during separate MRI and CT acquisitions. Final alignment quality was visually verified and adjusted using ITK-SNAP. This resampling step provided consistent image sizes, facilitating accurate comparisons during model training. To focus on anatomical regions and im-

prove sCT generation, binary masks were created using Otsu's thresholding method, which segmented the head region and excluded irrelevant areas¹³. Morphological operations, such as filling small holes and removing noise, were then applied to refine these masks. Lastly, CT images were intensity-thresholded and segmented to align with the head region, providing uniform, high-quality data for training the deep learning models.

Deep Learning Models for Pseudo-CT Generation

This study employed multiple deep learning models to generate pseudo-CT images, including U-Net, Pix2Pix, CycleGAN, and cGAN architectures. Each model was trained using a batch size of 2 due to hardware constraints (NVIDIA RTX 2080 GPU, 16 GB VRAM). The learning rate was set at 0.0002 and optimized using the Adam optimizer with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. U-Net and Pix2Pix models were trained for 200 epochs, while CycleGAN and cGAN were trained for 300 epochs to allow for better convergence. U-Net utilized MSE as its loss function, whereas Pix2Pix and cGAN employed a combination of adversarial loss and MAE. CycleGAN used adversarial loss, cycle consistency loss, and identity loss. PatchGAN discriminators were employed for GAN-based models.

U-Net Architecture: The U-Net model is a widely used convolutional neural network (CNN) designed primarily for image segmentation but also highly effective for image synthesis tasks, such as MRI-to-CT translation¹⁴. U-Net features a unique encoder-decoder structure with skip connections that link corresponding layers in the encoder (contracting path) and decoder (expanding path). The encoder path progressively reduces spatial dimensions while extracting features through convolutional layers, making the network more robust to variations in input data¹⁵. The decoder path, conversely, upsamples the feature maps to recover the spatial resolution. The skip connections play a crucial role by preserving high-resolution features from the encoder and merging them with upsampled layers in the decoder¹⁵. This design helps maintain detailed anatomical structures, which is essential for medical imaging tasks. For pseudo-CT generation, the U-Net model was trained on paired MRI-CT datasets, using Mean Squared Error (MSE) as the primary loss function, which measures the squared differences between predicted and actual CT images. Supplementary metrics like Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) were used to evaluate image fidelity, structural detail preservation, and noise reduction capabilities.

Pix2Pix Architecture: Pix2Pix is a conditional Generative Adversarial Network (cGAN) specifically designed for paired image-to-image translation tasks, making it highly suitable for generating synthetic CT (sCT) images from MRI scans⁶. Built on a U-Net-based encoder-decoder structure, Pix2Pix utilizes paired MRI-CT images for supervised training. The generator, based on U-Net, transforms MRI inputs into CT-like outputs, while the discriminator, a PatchGAN, assesses whether the generated CT images are authentic or synthetic.

The primary loss function for training Pix2Pix is Mean Absolute Error (MAE), which measures the average absolute differences between predicted and reference CT images, promoting higher accuracy. Additionally, metrics such as PSNR and SSIM were applied to quantify image quality and structural similarity, while Pearson Correlation Coefficient (PCC) evaluated the linear intensity correlation between generated and reference images. Pix2Pix's encoder-decoder structure with skip connections allows it to capture both global and fine-grained anatomical details, crucial for maintaining tissue structure and intensity fidelity in sCT images.

CycleGAN Architecture: CycleGAN, an unpaired image-to-image translation model, enables MRI-to-CT transformation without requiring paired datasets¹⁶. Unlike Pix2Pix, CycleGAN employs two generator-discriminator pairs: one set translates MRI images to CT, while the other translates CT images back to MRI. This bidirectional mapping is supported by cycle consistency loss, which ensures that an image translated to the opposite domain and then back to its original domain closely resembles the original input. CycleGAN also incorporates adversarial loss to encourage realistic CT outputs and identity loss to help preserve anatomical consistency. This architecture is particularly valuable when paired data is scarce or unavailable. By using cycle consistency loss, CycleGAN retains crucial anatomical details, making it suitable for medical imaging applications where alignment between MRI and CT domains is essential. Metrics such as SSIM and PCC were used to assess structural fidelity and intensity consistency, ensuring the generated images maintain a high degree of anatomical and intensity accuracy.

Conditional GAN (cGAN) Architecture: The conditional GAN (cGAN) model used in this study extends traditional GANs by conditioning the generation process on MRI input images, aiming to synthesize CT-like outputs. This model incorporates a PatchGAN discriminator, which evaluates images at the patch level, making it highly effective for capturing fine-grained texture details. The generator in the cGAN architecture follows an encoder-decoder structure, with 3D convolutions employed to capture volumetric features inherent in MRI and CT data. This design allows the cGAN to generate 3D synthetic CT volumes that align well with the spatial characteristics of MRI inputs. Key metrics for evaluating this model include MAE, MSE, SSIM, PSNR, and PCC. MAE and MSE provide insights into intensity accuracy, while SSIM and PSNR assess structural detail preservation and noise suppression. PCC evaluates the correlation in pixel intensities between synthetic and real CT images, further validating image quality and consistency. The 3D convolutional layers in the cGAN model are particularly beneficial in medical imaging, as they enable the model to capture depth information across slices, leading to more accurate and realistic 3D synthetic CT images.

Evaluation Metrics

To evaluate model performance, five metrics were employed: Mean Absolute Error (MAE), which measures the average absolute differences between real and synthetic CT values; Mean Squared Error (MSE), which emphasizes larger errors by squaring the differences between actual and generated pixels; Peak Signal-to-Noise Ratio (PSNR), which assesses image quality in terms of signal relative to noise; Structural Similarity Index (SSIM), which evaluates the structural fidelity between synthetic and real CT images; and Pearson Correlation Coefficient (PCC), which quantifies the linear correlation between pixel intensities in the original and generated images.

Quantitative and Statistical Analysis

Quantitative analysis compared the performance of U-Net, Pix2Pix, CycleGAN, and cGAN models by averaging MAE, MSE, SSIM, PCC, and PSNR values. Statistical validation used the Wilcoxon signed-rank test to compare the Pix2Pix model against alternative models, with significance set at $p < 0.05$.

Results

This section presents the outcomes of applying deep learning algorithms to generate synthetic CT (sCT) images from MRI data for radiotherapy dose planning, supported by visualizations and quantitative analyses.

Preprocessing Pipeline for Image Quality Enhancement

The preprocessing pipeline, including morphological transformations, enhanced the segmentation accuracy of binary masks, as shown in **Figure 1**. Here, Panel A illustrates the initial binary masks, and Panel B shows refined masks after

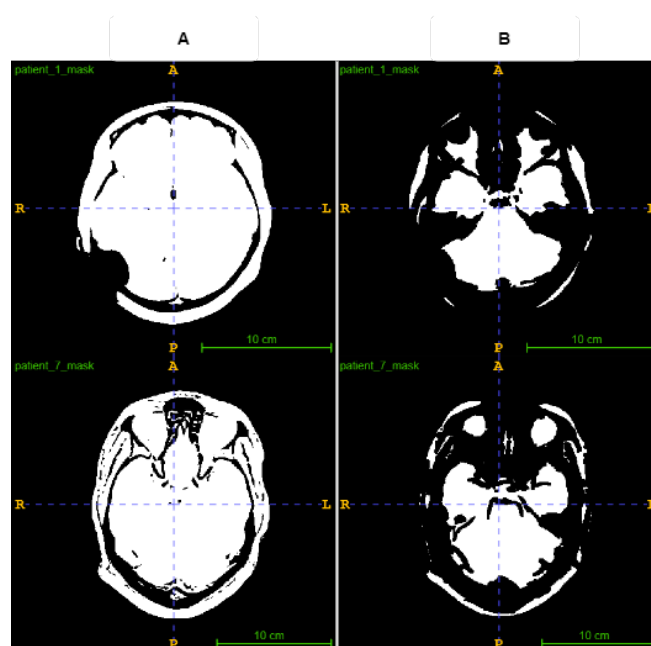


Figure 1. Results of Morphological Operations on Binary Masks.

applying the Binary Fill Hole Algorithm and Binary Erosion, improving segmentation fidelity for subsequent analyses.

Enhanced preprocessing for MRI and CT images is demonstrated in **Figure 2**, where Panel A shows the processed CT image after intensity thresholding, morphological closing, and masking resulting in improved anatomical clarity and background standardization. Panel B shows the original CT image, which contains non-anatomical background elements and higher noise levels prior to preprocessing.

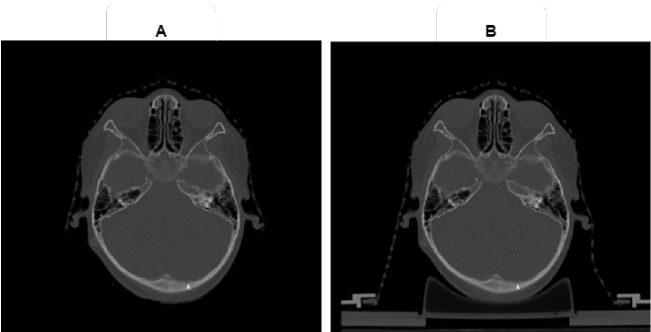


Figure 2. Effect of Noise Reduction and Preprocessing on CT Images.

Pseudo CT Generation Performance Across Models

Various deep learning models (U-Net, Pix2Pix, CycleGAN, and cGAN) were employed to generate sCT images from MRI data. Training and validation loss curves for each model (**Figure 3**) show smooth convergence in U-Net and CycleGAN, with Pix2Pix achieving rapid loss reduction but exhibiting slight fluctuations. cGAN converged the fastest, indicating robust learning across all models.

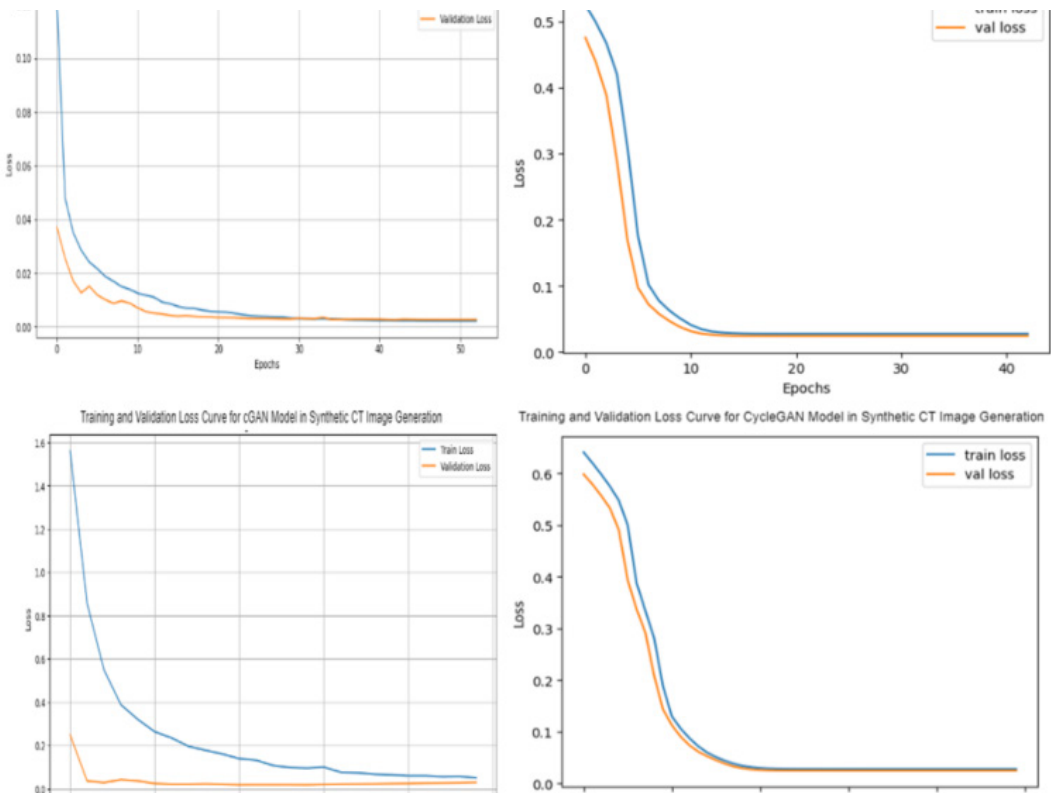


Figure 3. Training and Validation Loss Curve for all the Models in Synthetic CT Image Generation.

Quantitative Evaluation of Synthetic CT Quality

The quantitative results presented in Figures 4–8 and **Table 1** were computed on the test set comprising 2 subjects, which were not included in the training or validation stages. Metric values reflect the mean computed across all axial slices of the test subjects. To assess image quality, five key metrics were applied: PSNR, SSIM, MAE, MSE, and PCC.

PSNR: The comparison of PSNR values across models is illustrated in **Figure 4**, where Pix2Pix achieved the highest PSNR, indicating superior image fidelity and reduced signal loss. CycleGAN and cGAN also performed well, while U-Net showed lower PSNR, suggesting challenges in detail preservation.

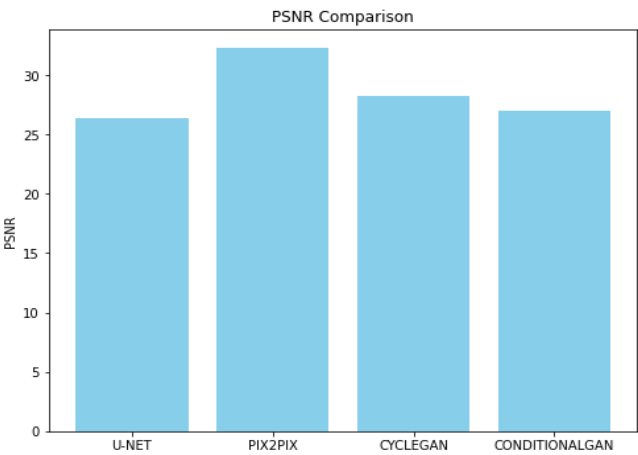


Figure 4. PSNR Comparison across Deep Learning Models for Synthetic CT Image Generation.

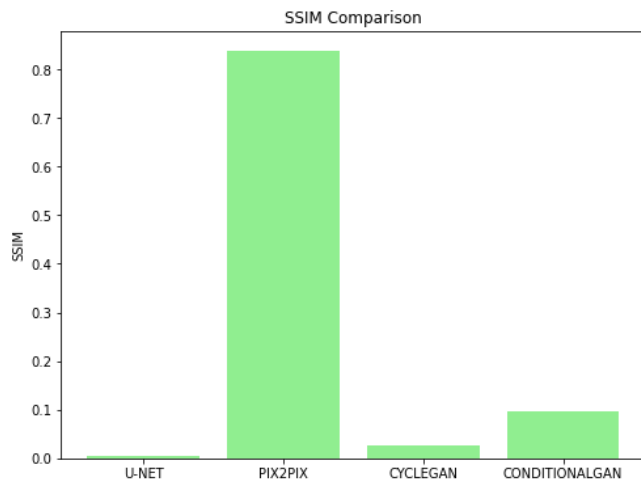


Figure 5. SSIM Comparison across Deep Learning Models for Synthetic CT Image Generation.

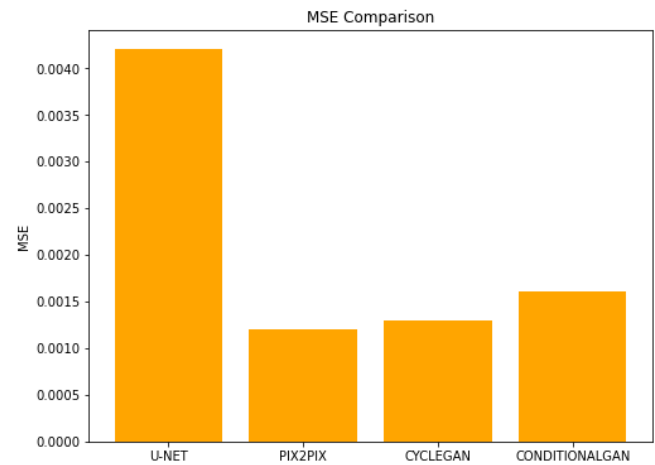


Figure 7. MSE Comparison across Deep Learning Models for Synthetic CT Image Generation.

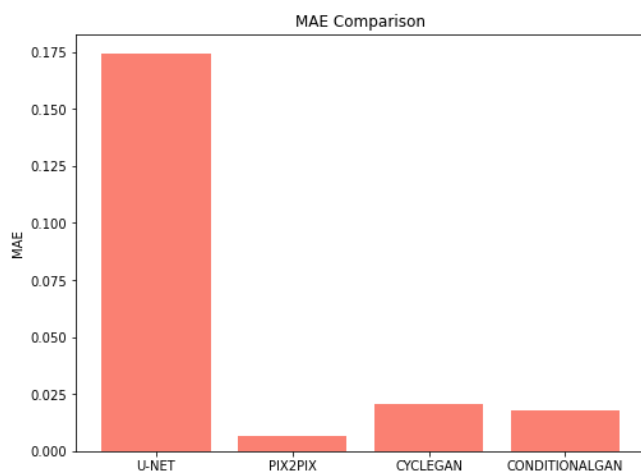


Figure 6. MAE Comparison across Deep Learning Models for Synthetic CT Image Generation.

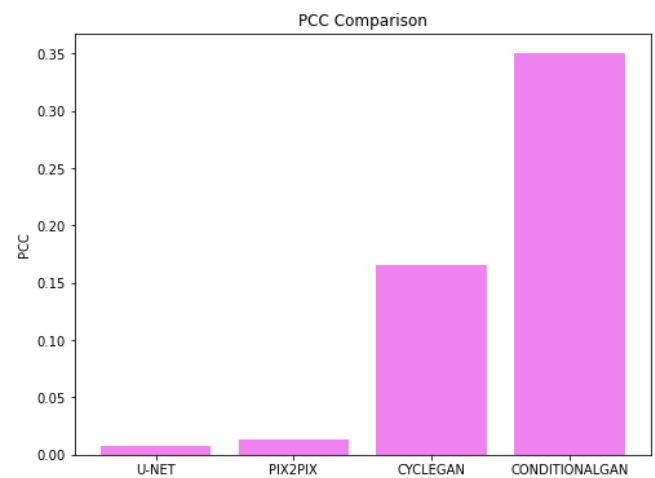


Figure 8. PCC Comparison across Deep Learning Models for Synthetic CT Image Generation.

- SSIM:** As shown in **Figure 5**, Pix2Pix attained the highest SSIM score, reflecting its ability to retain structural similarity. In contrast, U-Net, CycleGAN, and cGAN produced lower SSIM values, indicating limitations in structural fidelity.
- MAE:** **Figure 6** presents the MAE values, with Pix2Pix achieving the lowest error, reflecting high accuracy. CycleGAN and cGAN performed similarly well, while U-Net had the highest MAE, indicating substantial room for improvement.
- MSE:** The comparison in **Figure 7** shows Pix2Pix with the lowest MSE, followed closely by CycleGAN and cGAN. U-Net exhibited the highest MSE, indicating larger errors in synthetic image generation.
- PCC:** **Figure 8** illustrates PCC values, with cGAN achieving the highest score, demonstrating strong linear intensity correlation with reference CT images, followed by CycleGAN. Pix2Pix and U-Net had relatively lower PCC values, suggesting potential limitations in intensity alignment.

Statistical Analysis

Table 1 provides the p-values from the Wilcoxon signed-rank test comparing Pix2Pix with other models. Significant differ-

ences ($p < 0.05$) were found in most metrics, indicating Pix2Pix's superior performance, although similarities were noted in MSE for CycleGAN and cGAN.

Table 1. shows the p-values from the Wilcoxon signed-rank test comparing Pix2Pix with U-Net, CycleGAN, and cGAN to identify statistically significant differences across performance measures.

MODEL	U-NET	PIX2PIX	CYCLEGAN	CONDITIONALGAN (cGAN)
PSNR	7.89×10^{-161}	reference	1.50×10^{-114}	1.69×10^{-158}
SSIM	3.49×10^{-165}	reference	3.37×10^{-165}	6.73×10^{-165}
MAE	3.33×10^{-165}	reference	5.05×10^{-12}	1.84×10^{-8}
MSE	3.33×10^{-165}	reference	9.03×10^{-1}	4.79×10^{-1}
PCC	6.51×10^{-2}	reference	1.68×10^{-145}	6.06×10^{-165}

Discussion

In this study, we explored the performance of four deep learning models (U-Net, Pix2Pix, CycleGAN, and cGAN) for generating synthetic CT (sCT) images from MRI data. We evalu-

ated each model based on key metrics such as PSNR, SSIM, MAE, MSE, and PCC to assess their image quality, structural fidelity, and error minimization capabilities. The following discussion compares our results with previously published studies, highlighting strengths and areas for improvement in each model.

Our models produced PSNR values of 26.353 for U-Net, 32.2792 for Pix2Pix, 28.244 for CycleGAN, and 27.0272 for cGAN. These values show significant improvement, with Pix2Pix achieving a PSNR comparable to studies like Woltcrink, Dinkla, Savenije, Seevinck, van den Berg and Išgum⁷, which reported a PSNR of 32.5 ± 0.7 , and surpassing that of Nie, Trullo, Lian, Petitjean, Ruan, Wang and Shen⁸, which recorded 27.6 ± 1.3 and Ranjan, Lalwani and Misra⁶ which recorded 21.422 ± 3.964 . The higher PSNR values, particularly for Pix2Pix, indicate a better signal-to-noise ratio, meaning that the synthetic CT images generated by our models contain less noise and greater clarity. Although the Pix2Pix model performed best in noise reduction, there remains room for optimization, particularly in CycleGAN and cGAN, to reach even higher PSNR levels. The promising PSNR values across all models suggest that our models are capable of producing high-quality reconstructions. However, incorporating further noise reduction techniques or refining the architectural parameters such as adding residual blocks as used in studies like Ranjan, Lalwani and Misra⁶ could help enhance output clarity across all models.

SSIM scores indicate how well the models retain structural features between synthetic and target images. Among our models, Pix2Pix achieved the highest SSIM (0.8378), approaching the structural fidelity observed in studies such as Maspero, Bentvelzen, Savenije, Guerreiro, Seravalli, Janssens, van den Berg and Philippens¹⁷ with an SSIM of 0.86 ± 0.03 . Conversely, U-Net (0.0048) and CycleGAN (0.0253) showed considerably lower SSIM scores, suggesting limited structural similarity and potential loss of anatomical details in their reconstructions. This disparity in SSIM values indicates that Pix2Pix is the most successful in maintaining structural integrity, while U-Net and CycleGAN may require architectural modifications or enhanced training methods, such as multi-scale feature extraction, to improve structural accuracy.

The MAE results reveal considerable accuracy in error minimization across our models. Our models achieved MAE values of 0.1741 for U-Net, 0.0065 for Pix2Pix, 0.0209 for CycleGAN, and 0.0177 for cGAN, which are significantly lower than those found in other studies. For example, Han¹⁸ reported an MAE of 84.8 ± 17.3 , and Kazemifar, McGuire, Timmerman, Wardak, Nguyen, Park, Jiang and Owringi¹⁹ reported 47.2 ± 11.0 , both of which are markedly higher than our results. The low MAE values, particularly for Pix2Pix, suggest that our models are highly effective in minimizing absolute error between synthetic and target images, potentially contributing to more accurate clinical reconstructions. This performance in error reduction is encouraging, as low MAE values are critical for ensuring that sCT images closely resemble actual CT scans,

minimizing discrepancies that could affect diagnostic and treatment planning applications.

Similar to MAE, MSE provides insights into the squared differences between predicted and actual values, with lower values indicating better reconstruction fidelity. Our models produced MSE scores of 0.6542 for U-Net, 0.0012 for Pix2Pix, 0.0013 for CycleGAN, and 0.0016 for cGAN, which are significantly lower than the MSE reported by Han¹⁸ at 188.6 ± 33.7 . These low MSE scores reflect a reduction in error magnitude, affirming the high reconstruction accuracy of our models, particularly Pix2Pix, CycleGAN, and cGAN. This enhanced performance suggests that our models successfully minimize large errors, improving the consistency of the synthetic CT outputs. This result is particularly promising for applications where precise tissue differentiation is required, as lower MSE values indicate fewer extreme deviations in pixel intensity.

PCC evaluates the linear correlation between synthetic and target images, with higher values reflecting greater alignment. Our models yielded PCC values of 0.0081 for U-Net, 0.0130 for Pix2Pix, 0.1658 for CycleGAN, and 0.3501 for cGAN. Although lower than the NCC (Normalized Cross-Correlation) results reported in studies like Pan, Abouei, Wynne, Wang, Qiu, Li, Chang, Peng, Roper and Patel²⁰ at 0.983 ± 0.004 , cGAN's PCC score of 0.3501 demonstrates relatively better correlation performance within our models, indicating a stronger alignment with the target image in this model. Given the importance of PCC for accurate anatomical alignment, improving PCC across models, particularly U-Net would further enhance the reliability of sCT images. Implementing techniques to improve spatial correspondence, such as enhanced regularization during training, could be beneficial for increasing PCC values.

Conclusion

Among the models evaluated, Pix2Pix demonstrates the best balance across metrics, achieving relatively high SSIM, low MAE, and low MSE, indicating strong structural fidelity and error minimization. This model's performance makes it a promising candidate for clinical applications where both structural detail and low error are critical. However, its PSNR and PCC scores suggest there is room for improvement in noise reduction and anatomical alignment, which could be targeted in future model iterations. CycleGAN and cGAN also show competitive performance, particularly in PCC, suggesting they may excel in applications requiring greater alignment accuracy, though their SSIM and PSNR scores suggest a need for improved structural retention. Adjusting the loss functions or incorporating feature-matching layers could potentially enhance their structural similarity performance. In contrast, U-Net showed lower SSIM and PCC, suggesting that while it offers a basic reconstruction, it may not be as suitable for high-fidelity imaging tasks without further tuning. Enhancing U-Net's architecture to better capture multi-scale features could help improve its SSIM and PCC, making

it a more viable option for applications requiring detailed structural fidelity.

This study demonstrates that Pix2Pix holds particular promise for synthetic CT generation, with excellent error minimization and structural similarity. Meanwhile, CycleGAN and cGAN show potential for applications prioritizing correlation alignment. The overall performance of these models, with results comparable to or exceeding those of existing studies, underscores the effectiveness of deep learning approaches in medical image synthesis. Future work could focus on optimizing noise reduction and enhancing structural fidelity across all models, with a particular emphasis on improving U-Net's and cGAN's alignment metrics to ensure they meet clinical standards.

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